

Fertility, Housing, and Population Dynamics

From a stationary closure to a dated transition

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Equilibrium closure and transition dynamics

Last time: discussion of how to close population and housing markets:

- closed versus open population and the role of migration;
- stationary equilibrium versus a dated transition;
- fertility as a source of housing demand.

This time: consider a closed economy that begins in a steady state and experiences a permanent fall in the preference for children:

- fertility and the house price respond on impact;
- cohort sizes then change population and housing demand;
- the path may converge to a new steady state;
- we compare different versions in the quantitative model.

A toy economy

To fix ideas, consider a simplified economy with two household generations:

- At date t , there are Y_t young households and O_t old households; total population is $S_t = Y_t + O_t$.
- A household lives for two periods. A young household becomes old next period; an old household exits after the current period.
- Young households choose consumption, adult housing, and fertility n_t .
- Old households choose consumption and housing but not fertility.
- Both generations purchase housing services in one competitive market at price p_t .

Household problems

- A young household earns y_Y and chooses consumption c_Y , adult housing h^A , and fertility n :

$$\max_{c_Y, h^A, n} \{c_Y + \alpha \log h^A + v \log n\}, \quad c_Y + ph^A + [q + a(p + \omega)]n = y_Y.$$

- ω is the housing cost of children.
- Implied fertility and young-household housing demand:

$$n(p; v) = \frac{v}{q + a(p + \omega)}, \quad h_Y(p; v) = \frac{\alpha}{p} + an(p; v).$$

- An old household chooses consumption and housing:

$$\max_{c_O, h_O} \{c_O + \bar{h}_O \log h_O\}, \quad c_O + ph_O = y_O,$$

which gives $h_O(p) = \bar{h}_O/p$.

A higher housing cost lowers fertility and housing demand.

Competitive equilibrium

Housing is supplied competitively according to

$$H^s(p) = \bar{H}p^\varepsilon,$$

where $\bar{H}$ is the supply scale and ε is the price elasticity.

Given (Y_0, O_0) and a child preference v , an equilibrium is a path

$$\{p_t, c_{Y,t}, h_t^A, n_t, c_{O,t}, h_{O,t}, Y_t, O_t\}_{t \geq 0}$$

of housing costs, household choices, and cohort masses satisfying

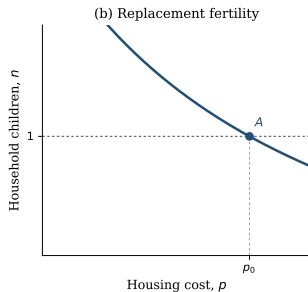
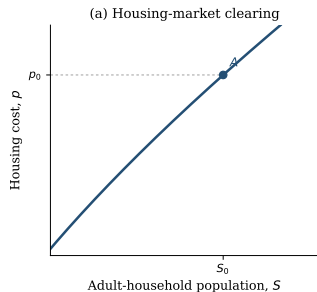
$$Y_t h_Y(p_t; v) + O_t h_O(p_t) = H^s(p_t),$$

and

$$Y_{t+1} = n(p_t; v)Y_t, \quad O_{t+1} = Y_t.$$

Current cohort masses determine the market-clearing price; the price determines fertility and therefore the next cohort.

Initial steady state



A steady state requires

$$n(p^*) = 1, \quad Y^* = O^*.$$

Here $n = 1$ is replacement fertility: each young household replaces itself next period.

At replacement fertility,

$$p_0 = (\nu - q)/a - \omega.$$

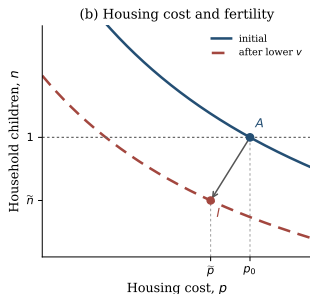
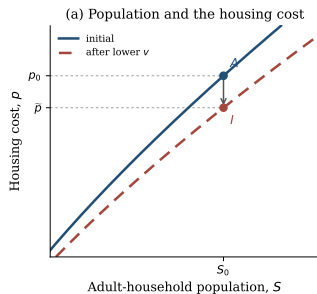
Housing clearing implies steady-state total population $S^* = Y^* + O^*$:

$$S^*(p) = \frac{2\bar{H}p^{1+\varepsilon}}{\alpha + \bar{h}_O + ap}.$$

$S^*(p)$ is increasing in p for $\varepsilon \geq 0$; point A is the initial steady state.

Impact response

Now suppose the preference for children falls permanently. Fertility and child-related housing demand fall before cohort masses adjust.

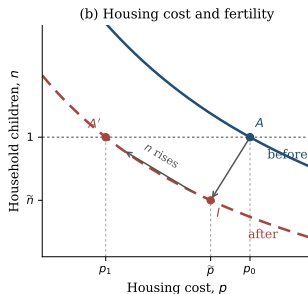
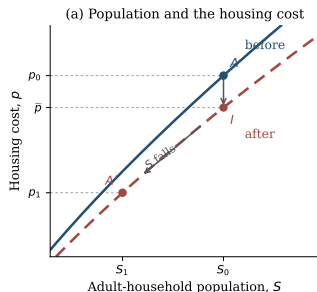


- **Shock:** $v_0 \rightarrow v_1 < v_0$.
- **Inherited cohorts:** $Y_t = Y_0$, $O_t = O_0$, $S_t = S_0$.
- **Housing cost:** lower child-related demand gives $p_0 \rightarrow \tilde{p}$.
- **Fertility:** $\tilde{n} = n(\tilde{p}; v_1) < 1$.

The economy moves from A to I : fertility falls below replacement, but population has not yet adjusted.

Demographic adjustment

► Dynamic path



- **Below replacement at I :**

$$Y_{t+1} = \tilde{n}Y_t < Y_t.$$

- **Cohorts:** smaller young cohorts eventually reduce S .
- **Housing cost:** lower demand moves $\tilde{p} \rightarrow p_1$.
- **Fertility:** lower p raises n along the red curve.
- **Point A' :** $n = 1$, $p_1 < p_0$, and $S_1 < S_0$.

Falling housing costs partly offset the initial fertility decline during the transition.

Below-replacement fertility and population dynamics

Define the old-to-young cohort ratio as $\mu_t = O_t/Y_t$. Since

$$S_{t+1} - S_t = Y_t(n_t - \mu_t),$$

population rises if and only if $n_t > \mu_t$. In particular, population can rise even below replacement:

$$\mu_t < n_t < 1.$$

For example,

$$(Y_t, O_t, n_t) = (100, 70, 0.8) \longrightarrow (Y_{t+1}, O_{t+1}) = (80, 100),$$

so total population rises from $S_t = 170$ to $S_{t+1} = 180$.

At a fixed housing cost p_t , aggregate housing demand rises when

$$D_{t+1}(p_t) > D_t(p_t) \iff (1 - \mu_t)h_O(p_t) > (1 - n_t)h_Y(p_t).$$

Cohort composition can sustain short-run population and housing-demand growth below replacement.

Quantitative exercise

The quantitative model adds sequential fertility, tenure choice, housing size, wealth accumulation, income risk, and bequests.

- Estimate household parameters to match 2023 fertility, housing, tenure, and wealth moments.
- Take the fitted joint distribution and inherited birth pipeline as the initial condition.
- Hold preferences fixed and simulate housing prices, fertility, tenure, and cohort sizes forward.

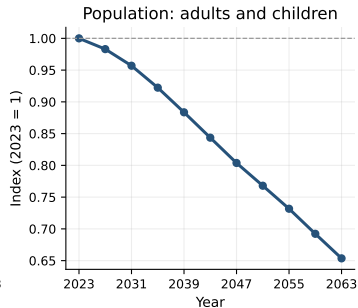
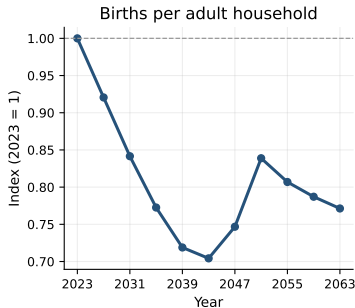
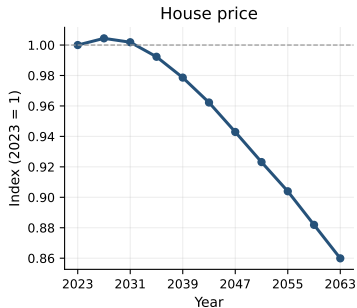
2023 calibration

Moment	Target	Model
Completed fertility	1.918	1.918
Childless share	0.188	0.186
Mean age at first birth	26.045	26.327
Share of first births at age 30+	0.260	0.242
Rooms response around first birth	0.720	0.380
Rooms gap: 3+ versus 1–2 children	0.368	0.359
Ownership gap: family versus no family	0.168	0.177
Ownership rate	0.575	0.546
Mean occupied rooms	5.780	6.710
Wealth / annual labor earnings	6.873	6.769
Annual bequests / aggregate wealth	0.0088	0.0086
Old-age wealth p90/p50	3.448	3.396

Largest discrepancies: the rooms response around first birth and mean occupied rooms.

The fitted 2023 economy going forward

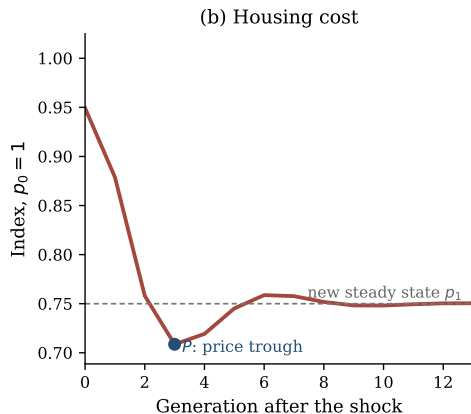
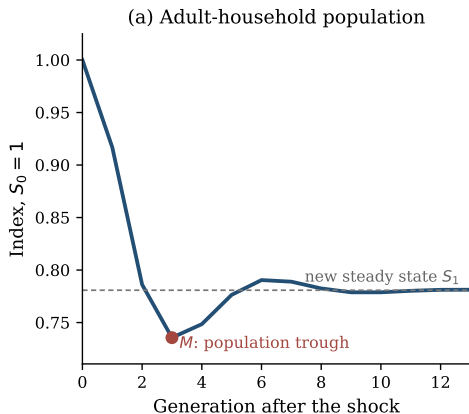
Starting from the fitted 2023 distribution, hold the economic environment fixed and follow the closed national economy forward.



Population counts one adult household unit plus the dependent children represented in that household.

The child stock contracts immediately; adult household formation delays the decline in housing demand and house prices.

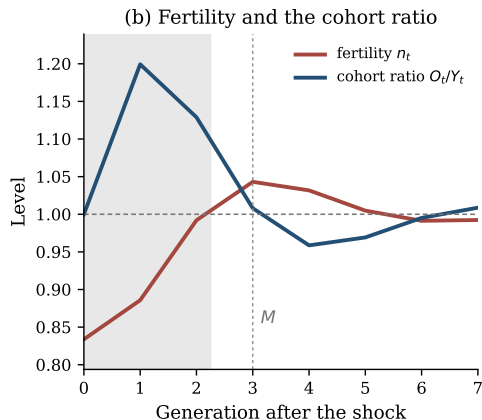
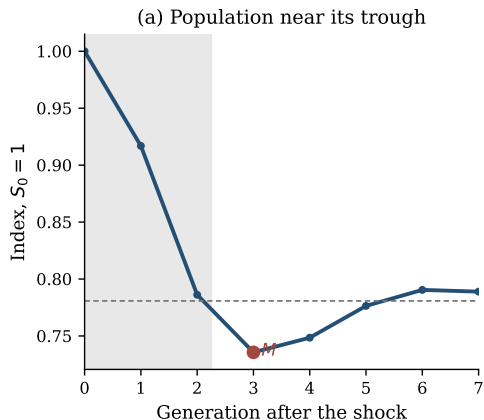
Population and housing-cost dynamics



Both population and the housing cost undershoot their new steady-state levels before recovering in this illustration.

The two-cohort illustration converges through progressively smaller oscillations around the new steady state.
Illustrative parameterization.

Cohort composition and the population trough



Population falls while $n_t < O_t/Y_t$ and turns upward when fertility overtakes the cohort ratio.

◀ Equilibrium diagram